

1 CASE STUDY 1: Reducing Abrasion Debris in Car Tyre Compound Design (ASTON MARTIN F1)

1.0.1 What types of abrasion take place in practice?

- Smear and abrasive wear

1.0.2 What is the difference between smear wear and abrasive wear in tyres?

- Smear wear occurs when the rubber is softened and smeared across the surface due to high friction and heat, rather than being cleanly cut or torn away. This leads to no abrasion pattern (smearing)
- Abrasive wear happens when hard particles or rough surfaces (e.g., road grit, asphalt texture) physically cut or tear pieces off the rubber tread. This is mainly common in high-performance vehicles.

1.0.3 Which wear process is best for tyre life?

- These are not the best for the tyre life, but degradation for abrasive wear is slower and more predictable.

1.0.4 How are these behaviors measured using lab tests?

- We could use electron microscopy due to its high resolution and magnification to observe micro-cracks, smear layers, and abrasion grooves
- To identify Abrasive wear - look out for scratches, particle pull-outs
- For smear wear - smooth, plastically deformed regions and a melted appearance.

1.0.5 How does the impact of this relate to rolling resistance and wet grip?

The USA market mode estimated that a peak of wear energy is at 13-17 J/m³ for free rolling

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1.0.6 What should the ideal compound look like and what does each ingredient add?

An ideal tyre tread compound is generally a blend of Natural Rubber (NR) and Synthetic Rubbers such as Solution-Styrene Butadiene Rubber (SSBR), reinforced with a filler system and controlled protective additives. NR contributes to wear resistance, toughness, and fatigue life; SSBR enhances wet grip adhesion. Silica, coupled with a silane agent, is added to SSBR to maximise wet traction and reduce hysteresis; it forms a reinforcing network that aids rolling resistance. Carbon Black can also be added in smaller quantities as a filler to improve abrasion resistance. Antioxidants protect against oxidative aging and ozone cracking.

1.0.7 If you use a blend of compounds how should the materials appear in their microstructure and why?

1.0.8 How are tyres made?

Compounding- Blends natural and synthetic rubbers with fillers and protective additives (antioxidants). The mixture is milled on two-roll mills to disperse uniformly. Each tire component (tread, sidewall, etc) is shaped via calendaring. After inspection, the green tyre is loaded into a mold where vulcanization occurs under pressure. The tread pattern and sidewall markings are imprinted during this stage as the rubber crosslinks. After this, the tyre undergoes post-cure finishing and uniformity checks.

2 CASE STUDY 2: HTP fairing from the A320 (NCC AIRBUS)

2.0.1 What Does the Fairing Do in Service?

The lightweight HTP fairing serves multiple services, such as reducing drag, helping to maintain stability, resist conditions at altitudes of 42,000ft (min temp -57 C and min pressure 17 kPa) and protect the flap. Resist conditions on ground in all operating conditions globally.

2.0.2 How Does the L Section of the Fairing Connect to the Wing?

The L Section of the Fairing contains flanges through which bolts and rivets can go through.

2.0.3 How You Choose the Materials for the Fairing – What Are We Looking to Optimize?

- Must be lightweight, not too dense, so it doesn't consume lots of fuel •
- Reduce drag, smooth surface
Water/UV resistance

2.0.4 What Are the Requirements?

- The material chosen for the mount should be such that:
 - The mass of the (entire) final component is 3.40 kg
 - The deflection at design load is 1.66 mm.
 - The main material mass is 3.02 kg.
- Resulting requirements:
 - Low density (low mass, lightweight)
 - Smooth finish to reduce drag
 - water/UV/corrosion resistant
 - High strength and high stiffness (low deflection)
 - Low cost
 - High durability/ sustainability
 - Recognised safety/ pass regulation requirements

2.0.5 How Can We Increase the Flexure Stiffness Whilst Keeping the Mass Low?

- Design with large second moment of area
- Very lightweight foamed core
- Rigid smooth composite skins

2.0.6 Which Materials Are Best for This Application?

- M21E aero grade cf pre-preg:
 - must be brought from the freezer
 - specifically manufactured to meet strict requirements for use on aircraft bodies: including performance (great material index), recognised safety and regulation requirements (passed as it is a qualified material)
- Epoxy resin matrix (typical grade)
- Unidirectional carbon fibres
- Foam core: rPET (recycled)
- Sandwich panel structure with ply-oriented lay-up:
 - Tool 'on site' at GKN (hand lay-up for this secondary structure)
- Debulk with consumables Trim excess - overlay
- Carbon fiber is more favorable

2.0.7 How You Make Them?

- Autoclave cure (more than one) a sandwich panel structure layer (three flies laid with a foam core and three flies laid at the top).
 - Sandwich panel structure with ply oriented layup:
 - 1) Each skin has three layers at 0, 90 and 45 deg with the fibre direction
 - 2) Each layer is degassed by vacuum before next layer is applied
 - 3) Tool 'on site' at GKN used for hand lay-up for (this 3 secondary structure)
 - 4) Autoclave cure with internal vacuum applied also
 - 5) Trim excess (overlay is cut off)
- Debulk with consumables

2.0.8 How Might You Test Them?

FEA (Finite Element Analysis) - non destructive

2.0.9 How Can You Exploit Waste / Recycle Stream Materials to Make Them?

- You can maximize the retention of the value of waste fiber fabrics by retaining fiber length, this will help to improve sustainability
- Using Patches:
 - 1) Virgin material: purchased material enters system
 - 2) Primary component: virgin material used to manufacture primary component
 - 3) Offcuts: offcuts identified and recovered for further use
 - 4) Patches: patches nested and cut from offcut material
 - 5) Secondary component: cut patches used to manufacture secondary component (i.e. flap track fairing)
 - 6) Residual waste: residual waste reduces, potentially used t for alternative recovery

3 CASE STUDY 3: Spinal fusion interbody cage

Challenge:

- Design cage concept with adaptive modulus to allow the new bone that has grown into the pores of the cage to bear an increasing load over time
- 15-20 kg/cm² applied on anterior and posterior cage, cage has to bear that for at least about 6 months
- Wolff's law: bone becomes stronger in areas that experience more stresses and weaker in areas of reduced stress
- So the bone eventually needs to start taking the weight
- Minimise movement of the cage, bones don't heal when there is micromotion. Shear is bad but you want a small amount of axial to stimulate
- Rapidly aging population apparently so more likely to have disc issues

Design considerations:

- Patient safety:
 - Ability to bear load in spine until it is certain that new bone will have formed (cage must not collapse before new bone can hold weight)
 - Align temporal adaptive modulus with bone formation
- include design features that drive bone formation
 - Pore size
 - Bone on-growth (Ti allows bone growth but is not radiolucent, PEEK is hydrophobic?? but can be resolved with HA coating)
- ability to assess bone formation (radiolucency)
 - Must be able to see bone grow through it for check ups
 - Adaptive radiolucency, where in the first few months can't see through it but the radiolucency increases by the time the check up would come
- Must resist sterilisation which involves the device being exposed to gamma radiation, too difficult with therapeutic protein as can have adverse toxic effect
- Could use nanoscale Ti as a coating for PEEK as would allow to see through the titanium whilst still getting the benefits of cell integration

Patient safety:

- Apophyseal ring bears the most weight. Very strong cortical bone
- Centre of vertebral body contains the softer cancellous bone
- Best designs are biomimetic:
 - Softer resorbable parts of the cage should be on inside
 - Harder permanent parts should be on outside
- Polymer only cages lead to failures due to deformations and degradations

Shape and size of material

- Hollow titanium tubes allow bone growth. Stem cells like this environment, perhaps they sense that this is a pit that has been created by osteoclasts so they go to remodel and repair it. (osteoclast bad, osteoblast good)
- Changing sub micron surface features at a nanoscale can allow for osteoinduction where it wouldn't have been possible. Increased surface area allows proteins/osteoblasts to attach onto nooks and crevices.
- Calcium phosphate (ceramic based) material with tube shape with a concave shape inside and nanoscale surface features can allow bone growth (osteoinduction)

In detail design considerations:

- Nanoscale features on surface of cages (easy for Ti)
- Size and concave shape of pores and grains (3D printing)

- Permanent surface charge to attract and bind endogenous growth factors to the surface
- Piezoelectric materials (e.g. PVDF)

Material choice

- 2 materials to allow complementary aspects to overcome weakness of just one material:
 - 1) Ti or Si₃N₄ or Coated PEEK as permanent outer layer
 - PEEKs biggest issue is not that it is weak but that cells don't grow on it so it is harder to create right size pores. Market's moved away from PEEK.
 - Titanium better but more expensive
 - Carbon Fibre PEEK is rarely used but can't be 3D printed and not good for the body. Laser sintering there would be powder and fibres and melting them together causes sharp bits of carbon fibre, not good.
 - 2) Use of innovative resorbable materials for inner layer (Mg Alloys)

Possible structures and other

- Use of porosity and innovative lattice structures (e.g. TPMS lattices) to offset modulus issues with harder materials like Ti
- Ti being too hard is a disadvantage. Increasing porosity will allow the titanium to become softer (i.e. lower modulus).
- Coating to slow Mg degradation (as degradation likely to be too quick for new bone growth). Magnesium degradation releases unwanted gases. Too much magnesium is bad.
- Use of antimicrobial coatings, since bacteria in body implants creates a biofilm infection which can't be fought with any current systemic antibiotic:
 - Silver good for antimicrobial coating, minimal disadvantages if used in smaller quantities
 - Copper not good as mimics hypoxic signal leading to unregulated angiogenesis, and blood vessels are important for bone formation
- Slot in design for a personalised resorbable inner
 - Could have a fast, medium and slow resorbing inner, and the surgeon could assess the bone quality during surgery and select a more patient specific resorption profile vs one size fits all resorption rate. However this approach has supply challenges which lead to higher costs and delays. Customised 3D printed cages
 - Permanent outer with variable degradation rates based on the patient could be an option to match likely biological response of patient

Final design

- 3D printed Ti outer to sit on the Apophyseal ring
- Open, porous structure with concave pores and submicron surface features
- Slot-in resorbable inner made, at least in part, from a Mg alloy to allow personalisation at point of use (fast, medium and slow degradation, more magnesium means faster) likely to be coated (chitosan)
- In the very centre, a bone graft chamber for point of use in addition of bone graft substitutes (eg rhBMP-2)

Manufacturing

- 3D printed Titanium:
 - Regulator doesn't like new methods so use well established process
 - Devices only require bench testing unless a new method is used (then have to do clinical studies)

- Use selective laser sintering

Osteo conduction and induction

- Osteoconduction:
 - creating a passive scaffold that allows stem cells to migrate across
 - Growing from something that already has bone to something where there wasn't bone
 - There has to be a continuous connection with it
- Osteoinduction:
 - Ability to get bone to grow away from other bone
 - BMP implanted in muscle pouch will grow bone there even with no surrounding bone

What is better, from a resorbable point of view, Magnesium alloy or PLA?

- Magnesium alloys:
 - newer
 - there is some evidence that Mg upregulates or contributes to bone formation
 - Not as radio dense as titanium but not as good as PEEK or PLA
 - Bones and cells grow on them easier
- PLA:
 - been around longer
 - you can see more of it
 - Less radio dense
 - Difficulty is that you can have variable reabsorption depending on how much acid builds up. The more acidic it becomes the faster it degrades.
 - Bones and cells don't grow on them as easily

4 CASE STUDY 4: Materials Selection for a Ferritic / Austenitic Stainless Transition Weld with a Proposal for a Non-Destructive Testing Method to ensure Volumetric Weld Integrity

4.0.1 Material Properties of a Filler

-Metal

-Filler must contain the right balance of alloying elements, so that it's post-weld microstructure complements that of the base materials. This ensures that the welded region will resist corrosion and cracking. Thermal properties govern how the weld pool forms, how quickly it cools, and what phases develop in the final weld.

4.0.2 Material Selection Process

-Schaeffler Diagram

4.0.3 Best Welding Process

4.0.4 Non-Destructive Evaluation (NDE)

- Inspection techniques designed to detect, characterise, and size flaws in welds and surrounding material without impairing the component. A well-designed NDE strategy

will reliably find cracks, porosity before they propagate into critical failures, thereby safeguarding both safety and operational uptime.

4.0.5 Different Types of NDE Methods

- Ultrasonic Testing sends high-frequency sound waves into the weld and detects reflections from internal flaws, making it suitable for volumetric inspection. Radiographic testing uses X-rays or gamma rays, sensitive to surface and near-surface defects, induces localised currents, and measures variations caused by flaws.

4.0.6 How to Use the Schaeffler Diagram

- Helps predict the balance of stainless weld metals based on their chromium and nickel equivalents. By plotting the chemical composition of the weld deposit in terms of these equivalents, one can locate its position on the diagram to see where the microstructure will be fully austenitic, Ferritic, or a mixture.

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4.0.7 Types of Faults/Defects in Welding

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4.0.8 Ultrasonic Testing and How It Works

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4.0.9 What Defect Can Be Detected From What NDE Method

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5 CASE STUDY 5: Biomedical Materials for Genicular Artery Embolisation to Treat Pain from Knee Osteoarthritis

5.0.1 What is Genicular Artery Embolization (GAE) and How It Can Be Used to Treat Pain in Knee Osteoarthritis

Genicular Artery Embolization (GAE) is a minimally invasive technique used to reduce chronic knee pain caused by osteoarthritis. It involves selectively embolizing the genicular arteries to reduce inflammation by limiting blood flow to the inflamed joint tissues.

5.0.2 Key Requirements for Embolic Materials Suitable for GAE

The embolic material must be:

- Biocompatible
- Uniform in size
- Compressible for catheter delivery
- Radiopaque or easily visualized
- Isobouyant in delivery solution
- Either resorbable or permanent depending on treatment goals

5.0.3 Best Materials for GAE and Why

Tris-acryl gelatin microspheres are preferred due to:

- Precise size control
- High compressibility
- Excellent biocompatibility

- Easy integration of radiopaque agents

5.0.4 Importance of Microsphere Size and How They Are Manufactured

Particle size affects targeting accuracy and safety. Microspheres are typically made using:

- Emulsion or suspension polymerization
- Crosslinking of gelatin or PVA
- Spray-drying for controlled size distribution

5.0.5 Explain the compressibility and isobouyancy property of the material and its importance in this application

Compressibility enables catheter delivery.

Isobouyancy prevents sedimentation in delivery fluid. Both properties help ensure even, predictable distribution in the vasculature.

5.0.6 Radiopaque and Dyed Microspheres: Purpose, Advantages, and Issues

- Purpose: Enhance fluoroscopic visibility.
- Advantages: Improve precision, reduce off-target effects.
- Issues: Potential changes to density, toxicity, and behavior.

5.0.7 How to Make an Embolic Agent

Steps:

1. Crosslink gelatin to form the microsphere core.
2. Coat or embed with tris-acryl polymer.
3. Screen particles by size.
4. Add radiopaque agents (optional).
5. Sterilize and store under sterile conditions.

6 CASE STUDY 6: Damping masses on the camshaft of an F1 engine (Mercedes AMG F1)

6.0.1 What is a damper mass?

A damper mass is a mechanical component of a camshaft that reduces fatigue on components to ensure smoother transitions of the carousel.

6.0.2 How does it work?

Opposes the direction to motion as the carousel rotates clockwise the damper mass (green) will rotate anti-clockwise in the bore (Check the case study 6 image)

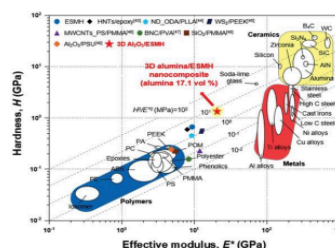
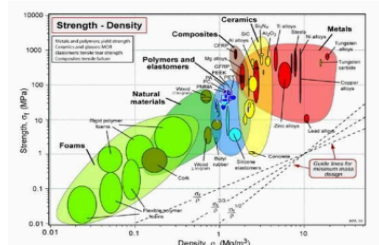
6.0.3 How should you choose the materials for the damper mass and the carousel that it sits in?

(Context) Coming into the 2010 season, some materials have been made illegal by FIA regulations. This includes our current material for the camshaft pendulum dampers. The material must now be below a density of 18,100 kg/m³. Packaging of the carousel is restricted to 18mm diametrically and you may fit as many pendulum dampers as you like per carousel.

Materials should obey the FIA regulations which states that:

- a) Magnesium based alloys;
 - b) Metal Matrix Composites;
 - c) Intermetallic materials;
 - d) Alloys containing more than 5% by weight of Beryllium, Iridium or Rhenium.
- Less than a density of 18,100 kg/m³.
 - Least wear on the damper masses or carousel
 - Match the damping torque and vibration order.
 - Rolling face interaction with EN9 steel. (Hardness = 150-350HV)
 - Operating in a power unit – exposed to temperatures up to 140 degC, oil, vibration within the carousel

Constraints	Objectives	Free Variables
Density < 18,100 kg/m ³ Material types i.e. must be a solid Hardness around EN9 steel (100 < HV < 400) Max Temperature > 140 degC Corrosion resistant to oil	Wear Resistance – high ((Hardness * Contact Area) / (Sliding Velocity * Load)) Hardness – high Density – high Yield Strength – high	Cost



6.0.4 What materials should you use and explain why?

- Tungsten Alloy
- Tungsten Alloys
- Tungsten Carbide
- Nickel Alloys Steel

(I put the table through chat to analyze this to help, the table was from the solutions, during the exam use your own words!!!)

Explanation:

The optimal material choice depends on the specific performance requirements and budget constraints of the application. **Steel** offers a balanced combination of properties, with a density of 7,000–8,000 kg/m³, hardness ranging from 100–800 HV, and yield strength of 200–500 MPa, all at a moderate cost of £100–200/kg—making it ideal for general-purpose engineering. For applications demanding **extreme hardness**, such as cutting tools or wear-resistant parts, **tungsten carbide** excels with a hardness of 1600–2500 HV, though its relatively low yield strength (300–350 MPa) and high cost (£200–300/kg) limit its versatility. **Tungsten alloys**, with a very high density of 17,000–18,500 kg/m³ and yield strength up to 1000 MPa, are best suited for high-load or high-impact uses like armor or counterweights, despite their high cost (£75–230/kg). On the more affordable end, **nickel alloys** cost only £40–75/kg and offer good corrosion resistance, but their lower yield strength (150–170 MPa) and hardness (100–400 HV) make them less suitable for structural or high-stress applications. Thus, the choice hinges on whether strength, hardness, weight, or cost is the top priority.

6.0.5 How might you make an individual damper mass?

Milling is the closest manufacturing option suitable for making an individual damper mass, making an assumption that the damper is a cylinder, it can be used for circular prismatics and it serves a tolerance of 0.02 - 0.5 mm.

6.0.6 How should we surface treat the damper mass to improve lifetime?

DLC coating

Nitriding

These processes avoid varying thermal expansions coinciding otherwise it will be a disaster

6.0.7 How do the vibration and torque calculations work?

-Vibration order must be around 4

- Damping torque must be around 19.67 Nm at 11000 rpm

$$\bullet \frac{m \times \omega_{engine} \times R \times 0.5 \times (R + e \frac{25}{r})}{1 + (\frac{25}{r})^2} \times n_{dampers}$$

$$\bullet \left(\frac{R}{e \times (1 + (\frac{37.5}{r^2}))} \right)^{1/2}$$

m = mass of damper
 ω_{engine} = engine speed
 R = carousel radius
 $e = p - r$
 r = roller radius
 p = bore radius
 $n_{dampers}$ = number of dampers (and bores)

6.0.8 How do you design to counter fretting / abrasion / wear in practice?

(From Cara Fox's solutions)

• Material Selection

- Use materials with higher wear resistance for the rolling/sliding components.
- Applying coatings such as chrome plating or specialized wear-resistant coatings can help reduce friction and protect surfaces from wear. For example, heat treatments, anodising or PVD/CVD (DLC - Diamond-Like Carbon) coatings can improve durability.

• Lubrication

- Ensure that proper lubrication is applied to the moving components. This helps reduce wear by minimizing direct contact between moving parts.
- So it doesn't have a bunch of friction between the the two surfaces that is rolling and non-rolling
- Avoid using too much lubrication because you will try to fight oil in
- Oil must be very light, low viscosity oil at the operating conditions otherwise you will get some damping effects associated with pumping the liquid around.

Important Note -Ensure you don't have tight tolerances (do not have a very small gap such that most of the damping mass covers the whole bore) the consequence of this is that if you have an oil in there you are squeezing that fluid as a film from one side to the other. If it has got a high viscosity its not going to allow the damper mass to rotate (Viscoelastic dissipative issues) therefore use a light oil and make the damper mass small.

• Friction Control

-Ensure that the damper is designed so there is enough clearance between components.
Surface finish of materials will make a huge difference, therefore machining methods should be considered for both carousel and damper mass.